Reasoning test

For admission to Class 11

About the test

- 30 questions in 30 minutes: 10 each of maths, English and science, mixed together.
- Half the maths and science questions are from the Class 10 syllabus (the chapters listed below); the rest test reasoning. Each question is marked Basics or Class 10.
- Taken online on a phone or computer, through a link sent to the parent's WhatsApp. The link works for 2 hours.
- Most questions have four options; a few ask for a number to be typed in.
- One mark for a correct answer. A quarter mark off for a wrong multiple-choice answer; nothing off for a wrong type-in answer or a blank.
- Answers can be changed before submitting. If the candidate leaves the page, the timer pauses.

What the test covers: basics

Basics questions test reasoning, not memory, using ideas taught up to Class 10. Chapters are from the NCERT (CBSE) books a student joining Class 11 in 2027 will have used; other boards teach the same ideas under their own names.

Maths

Class 6 · Mathematics

- *Whole Numbers* – number patterns and series
- *Playing with Numbers* – primes, factors, HCF and LCM
- *Fractions*
- *Mensuration* – perimeter and area
- *Ratio and Proportion*

Class 7 · Mathematics

- *Fractions and Decimals* – fractions
- *Data Handling* – averages (mean)
- *Simple Equations*
- *Comparing Quantities* – percentages, profit and loss, discounts, interest; ratio and proportion
- *Perimeter and Area*

Class 7 · Science

- *Motion and Time* – speed, distance and time

Class 8 · Mathematics

- *Linear Equations in One Variable* – simple equations
- *Squares and Square Roots* – squares and cubes
- *Cubes and Cube Roots* – squares and cubes
- *Comparing Quantities* – percentages, profit and loss, discounts, interest

Class 10 · Mathematics

- *Arithmetic Progressions* – number patterns and series
- *Probability*

Reasoning, not a textbook chapter: Clocks and calendars · Number codes and puzzles · Everyday units · Multi-step word problems · Time and work, pipes and tanks · Problems on ages · Mixtures · Overlapping groups (Venn diagrams)

English

Grammar, vocabulary and reasoning; nothing from the English reader: Vocabulary: meaning in context, opposites, idioms · Grammar: agreement, prepositions, tenses, linking words · Completing and ordering sentences · Analogies, odd one out, letter series and codes · Reading: inference, tone, assumptions, judging an argument · Logical deduction: statements, arrangements, family relations

Science

Class 6 · Science

- *Fun with Magnets* – magnets
- *Light, Shadows and Reflections* – light, shadows and plane mirrors
- *Electricity and Circuits* – electric circuits

Class 7 · Science

- *Heat* – heat and conductors
- *Acids, Bases and Salts* – acids, bases and indicators
- *Physical and Chemical Changes* – physical and chemical changes, rusting
- *Nutrition in Plants* – photosynthesis and plant respiration
- *Respiration in Organisms* – respiration; photosynthesis and plant respiration
- *Motion and Time* – speed
- *Light* – light, shadows and plane mirrors

Class 8 · Science

- *Force and Pressure* – force and pressure, pressure in liquids
- *Sound* – sound, and the speeds of light and sound

Class 9 · Science

- *Matter in Our Surroundings* – evaporation and condensation; change of state
- *Is Matter Around Us Pure?* – elements, compounds, mixtures and solutions
- *Force and Laws of Motion* – inertia, action and reaction
- *Gravitation* – mass and weight; floating, sinking and density
- *Work and Energy* – energy changes

Class 10 · Science

- *Electricity* – electric power and safety
- *Our Environment* – food chains

Reasoning, not a textbook chapter: Reading an experiment and fair tests · Applying a principle to a new situation

Class 10 syllabus questions

Five maths and five science questions come from these Class 10 chapters only: the first-term chapters, because the test can be taken before the year ends.

Maths

Class 10 · Mathematics

- *Ch 1. Real Numbers* – HCF and LCM, irrational numbers
- *Ch 2. Polynomials* – zeroes and coefficients
- *Ch 3. Pair of Linear Equations in Two Variables* – solving, word problems
- *Ch 4. Quadratic Equations* – roots, nature of roots, word problems
- *Ch 5. Arithmetic Progressions* – nth term, sum of terms
- *Ch 6. Triangles* – similarity, basic proportionality
- *Ch 7. Coordinate Geometry* – distance and section formulas
- *Ch 8. Introduction to Trigonometry* – trigonometric ratios

Science

Class 10 · Science

- *Ch 1. Chemical Reactions and Equations* – types of reactions
- *Ch 2. Acids, Bases and Salts* – pH, common salts, neutralisation
- *Ch 5. Life Processes* – nutrition, respiration, transport
- *Ch 6. Control and Coordination* – reflexes, hormones
- *Ch 9. Light – Reflection and Refraction* – mirrors, lenses, mirror formula, power
- *Ch 11. Electricity* – Ohm's law, resistances, power

Sample paper

These questions were written for this sheet. The real test has different questions, in the same style and at a similar level.

1. MATHS · BASICS
   Which of these numbers is divisible by both 3 and 4?

   A 132 **B** 138
   C 142 **D** 150

2. ENGLISH · BASICS
   The sentries stayed vigilant all night. As used here, "vigilant" most nearly means

   A tired **B** watchful
   C silent **D** brave

3. SCIENCE · BASICS
   A 1,000 W room heater is switched on for 3 hours. How many units (kilowatt-hours) of electricity does it use?

   Type-in answer: _____

4. MATHS · CLASS 10
   What is the HCF of 24 and 36?

   A 6 **B** 72
   C 4 **D** 12

5. ENGLISH · BASICS
   Choose the correct word: The cadets were proud ___ their achievement.

   A of **B** for
   C with **D** at

6. SCIENCE · CLASS 10
   Which gas is given off when zinc reacts with dilute hydrochloric acid?

   A Oxygen **B** Hydrogen
   C Carbon dioxide **D** Chlorine

7. MATHS · BASICS
   Kiran scored 54 marks out of 75 in a test. What percentage did Kiran score?

   Type-in answer: _____

8. ENGLISH · BASICS
   Which word is the odd one out?

   A Cricket **B** Hockey
   C Football **D** Chess

9. SCIENCE · BASICS
   A fuse is fitted in a household circuit so that, if too much current flows, it

   A raises the voltage to appliances **B** melts and breaks the circuit
   C stores the extra electricity **D** makes the bulbs glow brighter

10. MATHS · CLASS 10
    What is the 15th term of the arithmetic progression 5, 8, 11, …?

    Type-in answer: _____

11. ENGLISH · BASICS
Choose the word most nearly opposite in meaning to "humble".

A proud
B kind
C quiet
D poor

12. SCIENCE · CLASS 10
Which blood vessels carry blood away from the heart to the rest of the body?

A Arteries
B Veins
C Capillaries
D Nerves

13. MATHS · BASICS
A bus leaves at 9:40 am and travels 150 km at a steady 60 km/h. At what time does it arrive?

A 11:40 am
B 11:50 am
C 12:10 pm
D 12:30 pm

14. ENGLISH · BASICS
Choose the correct words: She ___ in this school since 2022.

A studies
B is studying
C has studied
D studied

15. SCIENCE · BASICS
Which of these is a compound?

A water
B oxygen
C iron
D air

16. MATHS · CLASS 10
What is the value of tan 45°?

A 0
B 1
C 1/√2
D √3

17. ENGLISH · BASICS
To "burn the midnight oil" means to

A spend money carelessly
B work late into the night
C light a lamp at midnight
D forget an important task

18. SCIENCE · CLASS 10
A 10 Ω resistor is connected across a 5 V battery. What current flows through it, in amperes?

Type-in answer: _____

19. MATHS · BASICS
After a 25% increase, a salary is ₹20,000 a month. What was it before the increase?

A ₹15,000
B ₹16,000
C ₹16,500
D ₹17,500

20. ENGLISH · BASICS
If COLD is written as DPME in a code, how is WARM written in the same code?

A VZQL
B XBSM
C XBSN
D YCTO

21. SCIENCE · BASICS
A 60 W bulb and a 1,500 W heater are each used for 2 hours. How many times more energy does the heater use?

A 2.5 times
B 12.5 times
C 25 times
D 50 times

22. MATHS · CLASS 10

What are the roots of the equation $x^2 - 5x + 6 = 0$?

A 2 and 3
B −2 and −3
C 1 and 6
D −1 and 6

23. ENGLISH · BASICS

"Carry a water bottle on the march; there are no taps on the route." This advice assumes that

A the marchers will need water
B taps on routes are often dirty
C the march will be a short one
D water bottles are not costly

24. SCIENCE · CLASS 10

A ray of light travels at an angle from air into glass. At the surface, the ray bends

A away from the normal
B along the surface
C towards the normal
D back into the air

25. MATHS · BASICS

Two fair dice are thrown together. What is the probability that the two numbers add up to 8?

A 1/6
B 1/9
C 7/36
D 5/36

26. ENGLISH · BASICS

All cadets wake at 5 am. Nobody who wakes at 5 am misses breakfast. Arun misses breakfast. Which statement must be true?

A Arun wakes at 5 am
B Arun is a cadet
C Some cadets miss breakfast
D Arun is not a cadet

27. SCIENCE · BASICS

A wooden block floats in water with exactly half of its volume under the surface. Water has a density of 1 g/cm³. What is the density of the wood?

A 1 g/cm³
B 2 g/cm³
C 0.25 g/cm³
D 0.5 g/cm³

28. MATHS · CLASS 10

The point (5, 4) lies exactly halfway between (2, a) and (b, 6). Find a + b.

Type-in answer: _____

29. ENGLISH · BASICS

P is taller than Q. R is shorter than Q. S is taller than P. Who is the second tallest?

A S
B Q
C P
D R

30. SCIENCE · CLASS 10

Three 6 Ω resistors are connected in parallel across a 12 V battery. What total current is drawn, in amperes?

Type-in answer: _____

Answers

1	**A. 132**	132 = 12 × 11, so it is divisible by both. 138 and 150 are divisible by 3 but not by 4; 142 is divisible by neither.
2	**B. watchful**	Vigilant means keeping careful watch for danger, which is exactly what a sentry does. Tired or silent sentries may still be vigilant.
3	**3**	1,000 W is 1 kW, and energy is power × time: 1 kW × 3 h = 3 kWh, which is 3 units. This is a type-in question.
4	**D. 12**	$24 = 2^3 \times 3$ and $36 = 2^2 \times 3^2$. The HCF takes the lowest power of each common prime: $2^2 \times 3 = 12$. 72 is the LCM.
5	**A. of**	"Proud" is followed by "of": proud of something. The other prepositions are not used with proud in this sense.
6	**B. Hydrogen**	A metal and a dilute acid give a salt and hydrogen: zinc chloride and hydrogen here. Carbon dioxide comes from acids with carbonates.
7	**72**	54 ÷ 75 = 0.72, which is 72%. Another way: 75 marks is 100%, so each mark is 4/3 %, and 54 × 4/3 = 72. This is a type-in question.
8	**D. Chess**	Cricket, hockey and football are outdoor team games played with a ball. Chess is a board game for two players.
9	**B. melts and breaks the circuit**	A fuse wire melts when the current is too large, which breaks the circuit and stops the flow before wires overheat or catch fire.
10	**47**	a = 5 and d = 3, so the 15th term is a + 14d = 5 + 42 = 47. This is a type-in question: the answer is typed as a number.
11	**A. proud**	Humble means modest, not thinking oneself important, so the opposite is proud. A humble person may also be kind or quiet, but those are not opposites.
12	**A. Arteries**	Arteries carry blood away from the heart under high pressure; veins bring it back. Capillaries link the two, and nerves carry signals, not blood.
13	**C. 12:10 pm**	The journey takes 150 ÷ 60 = 2.5 hours, which is 2 hours 30 minutes. 9:40 am plus 2 hours 30 minutes is 12:10 pm.
14	**C. has studied**	"Since 2022" marks an action that began in the past and continues now, which takes the present perfect: "has studied". The simple present and past do not fit "since".
15	**A. water**	Water is hydrogen and oxygen chemically joined, so it is a compound. Oxygen and iron are elements, and air is a mixture.
16	**B. 1**	In a right triangle with a 45° angle the two shorter sides are equal, so tan 45° = opposite ÷ adjacent = 1. $1/\sqrt{2}$ is sin 45° or cos 45°.
17	**B. work late into the night**	The idiom comes from working by lamplight after midnight: it means staying up late to work or study. Lighting the lamp is the picture, not the meaning.
18	**0.5**	By Ohm's law I = V ÷ R = 5 V ÷ 10 Ω = 0.5 A. This is a type-in question: the answer is typed as a number.

19	**B. ₹16,000**	The new salary is 125% of the old one, so the old salary is 20,000 ÷ 1.25 = ₹16,000. Taking 25% off ₹20,000 gives the wrong ₹15,000.
20	**C. XBSN**	Each letter moves one step forward: C→D, O→P, L→M, D→E. So W→X, A→B, R→S, M→N, giving XBSN.
21	**C. 25 times**	Both run for the same time, so energy goes up with power alone: 1,500 ÷ 60 = 25. The 2 hours cancels out.
22	**A. 2 and 3**	$x^2 - 5x + 6 = (x - 2)(x - 3)$, so the roots are 2 and 3. 1 and 6 multiply to 6 but add to 7, not 5.
23	**A. the marchers will need water**	The advice only makes sense if the marchers will need water along the way. The other statements are not needed for it to hold.
24	**C. towards the normal**	Light slows down in glass, a denser medium, so it bends towards the normal. Going from glass into air it bends away from the normal.
25	**D. 5/36**	There are 36 equally likely outcomes. A total of 8 comes from (2,6), (3,5), (4,4), (5,3) and (6,2), which is 5 outcomes: 5/36.
26	**D. Arun is not a cadet**	Every cadet wakes at 5 am, and everyone who does has breakfast. Arun does not, so Arun cannot wake at 5 am and so cannot be a cadet.
27	**D. 0.5 g/cm³**	A floating block pushes aside its own weight of water. It does so with half its volume, so it weighs as much as half its volume of water: 0.5 g/cm³.
28	**10**	Midpoint x: (2 + b)/2 = 5, so b = 8. Midpoint y: (a + 6)/2 = 4, so a = 2. Then a + b = 10. This is a type-in question.
29	**C. P**	Putting the statements together gives S > P > Q > R, so P is second tallest. S is the tallest and R the shortest.
30	**6**	In parallel 1/R = 3 × 1/6 = 1/2, so R = 2 Ω and I = 12 V ÷ 2 Ω = 6 A. Treating them in series (18 Ω) gives 0.67 A.